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VES-based characterization of lithological heterogeneity for potential groundwater development in the quaternary floodplain of Jashore Sadar, Bangladesh

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Abstract

Subsurface geology in deltaic floodplains is highly heterogeneous, strongly influencing groundwater development. In response to growing groundwater stress and the need for potential management, this study investigates the capability of vertical electrical sounding (VES) to characterize lithological variability and infer hydrostratigraphic conditions within the active floodplain of the Quaternary deposits of Jashore, Bangladesh. A total of 18 VES surveys were conducted using the Schlumberger array with investigation depths of 90–150 m. One-dimensional inversion modeling using IX1D was validated against nearby borehole logs, showing strong agreement for layer depth and thickness ($R^2 > 0.95$) and approximately 75% agreement in lithological classification. The results show five geoelectric layers with resistivity values ranging from 6 to 60 Ωm , corresponding to topsoil, clay, silty clay, and sandy lithologies, indicating considerable lateral and vertical heterogeneity across the floodplain. Furthermore, geo-electric cross-sections indicate that shallow or thick sandy units are concentrated around VES-3–7, 9–11, and 14–17, whereas clay-rich deposits dominate VES-1, 12, 13, and 18. In particular, the sandy layers characterized by relatively high resistivity (36–60 Ωm) and substantial thickness (32.35–94 m) are interpreted as potential aquifer zones, while the overlying clay and silty clay units suggest as confining to semi-confining layers. Moreover, the absence of very low resistivity signatures suggests limited saline water intrusion within the investigated depth range. Although hydrochemical and pumping-test data are still required for confirmation, our study develops a conceptual framework of lithological heterogeneity using VES for potential groundwater management applicable for similar floodplain environments.

Keywords Lithological heterogeneity, Vertical electrical sounding (VES), Groundwater exploration, Potential groundwater development, Jashore Sadar



1 Introduction

Groundwater plays a vital role in global freshwater supply and food security. At present, approximately 21–30% of total global freshwater withdrawals originate from groundwater sources. Its significance has increased steadily over time, with per-capita groundwater use rising from 124 m³ in 1950 to 152 m³ per person per year in 2021, representing an increase of about 22.6% [1]. Agriculture accounts for nearly 69% of total groundwater withdrawals worldwide, while municipal, industrial, and commercial sectors together account for the remaining 31%. This expanding dependence, coupled with climate variability and population growth, has placed considerable stress on aquifer systems globally [2–5].

Bangladesh is among the most groundwater-dependent countries in the world. Each year, approximately 32 km³ of groundwater is withdrawn, of which about 90% is used for irrigation and only 10% for domestic and industrial purposes combined [6]. National assessments that integrate arsenic contamination, salinity intrusion, and groundwater storage depletion indicate that between 5 and 24% of the land area and 4.5–17% of the population are exposed to high or extreme groundwater risk, affecting up to 24 million people [7]. These risks are particularly pronounced in the northwest and southwest regions, where declining groundwater levels and seasonal shortages have become increasingly evident [8]. The combined effects of intensive dry-season irrigation, variable monsoon rainfall, and limited surface water availability have intensified pressure on aquifer systems in these regions.

Within this regional setting, Jashore District in southwestern Bangladesh exhibits growing groundwater stress. At the local scale, especially in Jashore Sadar, groundwater challenges are characterized not by complete depletion but by spatial variability in aquifer productivity, seasonal water scarcity, and emerging water-quality concerns [9, 10]. Hydrogeological evidence suggests that moderate to good groundwater potential still exists in parts of the area; however, increasing agricultural abstraction and expanding urban demand are placing additional pressure on the resource. Experiences from surrounding districts and adjacent regions of India demonstrate that unmanaged over-extraction, pollution, and climate variability can rapidly transform moderately stressed aquifers into severely depleted systems [8, 11]. Therefore, understanding the local subsurface conditions is essential for potential groundwater management.

The hydrogeological complexity of Jashore Sadar is largely controlled by Quaternary floodplain deposits, which exhibit pronounced lithological heterogeneity [12]. Subsurface layers vary significantly in depth, thickness, and textural composition over short distances. Such heterogeneity directly influences aquifer properties, including transmissivity, storage capacity, and well yield. As groundwater levels decline during the dry season, irrigation wells in certain locations frequently run dry [13]. In addition, the area lies within one of the most arsenic-affected vulnerable zones in Bangladesh [13], further increasing risks to public health and agricultural sustainability. These combined challenges highlight the importance of detailed subsurface characterization at the local scale.

Although previous studies in southwestern Bangladesh have mainly focused on groundwater quality, hydrochemical facies, arsenic contamination, and associated health impacts [14–16], site-specific characterization of subsurface lithological heterogeneity and aquifer geometry in Jashore Sadar remains limited. In particular, the lateral continuity, thickness variability, and vertical distribution of potential aquifer units within the

Quaternary floodplain deposits are still poorly understood. This limitation is important because floodplain aquifers are highly heterogeneous, with rapid lateral facies changes that strongly influence groundwater occurrence, recharge, and well productivity.

Most previous investigations in the region have relied primarily on sparse and spatially discontinuous borehole data [17, 18], which provide only point-scale information and are often insufficient to capture the complex three-dimensional variability of alluvial sediments. As a result, groundwater development frequently proceeds with limited understanding of subsurface architecture, increasing the uncertainty of aquifer targeting, well placement, and potential abstraction planning.

To address this gap, the present study applies vertical electrical sounding (VES) to provide a more spatially continuous interpretation of subsurface lithological variability in the floodplain environment of Jashore Sadar. Unlike previous regional studies, this research integrates 18 VES soundings with borehole validation and quantitative uncertainty assessment to improve the reliability of geoelectrical interpretations. The study therefore provides new insight into the lateral and vertical heterogeneity of Quaternary floodplain deposits and their hydrostratigraphic implications for groundwater development. Specifically, the research aims to characterize subsurface lithological layers, delineate hydrostratigraphic units based on resistivity and thickness variations, and identify comparatively favorable zones for groundwater development. Overall, the study demonstrates the potential of VES as a cost-effective approach for reducing uncertainty in aquifer exploration and supporting evidence-based groundwater management in complex floodplain environments.

2 Study area and its geology and hydrogeology

To delineate effectively hydro-stratigraphical conditions of any area using geophysical methods, it is very crucial to know the local geological, geomorphological and hydrogeological conditions [19–21]. In fact, these factors not only influence subsurface resistivity responses but also control aquifer geometry, continuity, and groundwater flow behavior.

Jashore Sadar is an upazila of Jashore district which is located southwestern part of Bangladesh (Fig. 1). It is in between 23° 04' and 23° 20' north latitudes and in between 89° 06' and 89° 08' east longitudes and with an area of about 435.22 sq. km. Tectonically, the study area falls under the Faridpur trough tectonic element which is a part of the Bengal Basin [22].

Geomorphologically, the region belongs to the active Ganges deltaic floodplain, characterized by fluvio-deltaic processes dominated by meandering river systems, overbank flooding, natural levee formation, and abandoned channel deposits [17]. The presence of the Bhairab and Chitra rivers, along with numerous floodplain depressions and seasonal water bodies, indicates an active floodplain environment. Such environments typically exhibit rapid lateral and vertical changes in lithology due to shifting channel positions and varying depositional energy conditions.

Geologically, the study area belongs to the early Ganges delta, which is part of the Ganges deltaic plain. In fact, the area is underlain by Quaternary alluvial deposits, consisting predominantly of alternating sequences of fine to medium sand, silt, clay, and silty clay. These deposits show significant heterogeneity over short distances. The thickness of Quaternary sediments in the southwestern Bengal Basin commonly exceeds several tens to hundreds of meters, with localized variations depending on depositional settings and

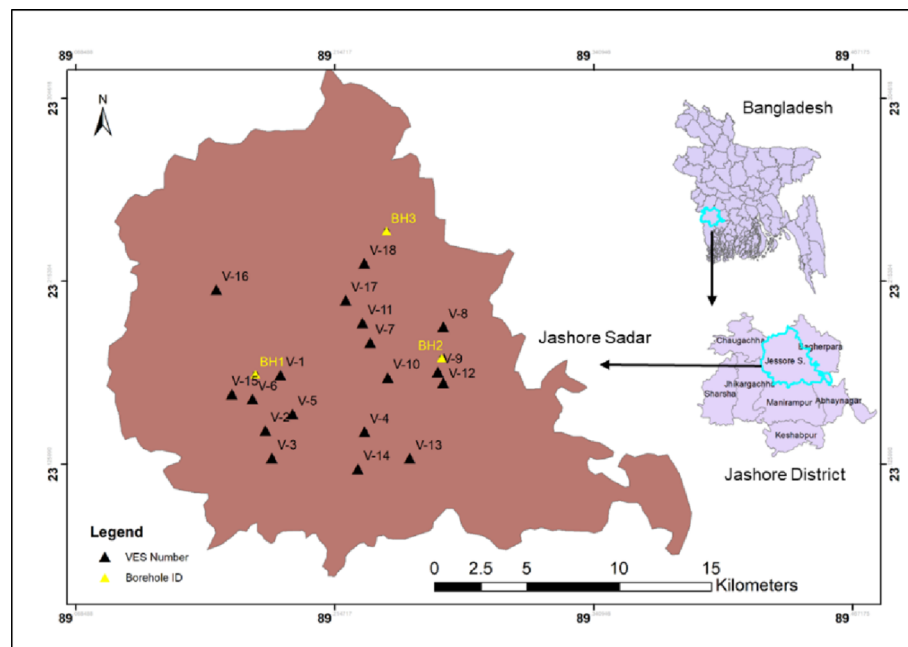


Fig. 1 Study area

basin subsidence. In floodplain environments such as Jashore, near-surface fine-grained deposits (clay and silt) typically overlie thicker sandy sequences at depth, forming multi-layered aquifer systems [23].

Stratigraphically, the area is generally described as comprising Pleistocene Madhupur Clay overlying Mio-Pleistocene Dupi Tila Formation. However, site-specific stratigraphic logs for Jashore Sadar are limited. Therefore, the stratigraphic profile presented in Fig. 2 from nearby Khulna is used as a regional reference framework rather than a direct representation of the study area. This approach is considered reasonable because both Khulna and Jashore Sadar lie within the same Ganges deltaic depositional system and Bengal Basin tectonic setting, and previous studies have reported broadly similar lithological sequences (sand–silt–clay alternations) across southwestern Bangladesh. Nevertheless, given the known lateral heterogeneity of floodplain environments, the Khulna stratigraphy is used here only to provide a generalized depositional context, and not to imply exact layer continuity or thickness in Jashore Sadar. The present study addresses this limitation by using VES data to derive site-specific subsurface layering, thereby complementing and refining the regional stratigraphic understanding.

Hydrogeologically, the Mio-Pleistocene sediments of the Dupi Tila Formation, which forms the aquifer systems, lies beneath the Pleistocene Madhupur Clay Formation [12]. The groundwater table of the study area varies from 3.94 to 9.7 m throughout the year [11]. Regional groundwater flow is generally directed from northwest to southeast, although local variations may occur due to pumping and surface water interactions. The groundwater quality varies from fresh to brackish and the dominant hydrogeochemical facies of the area are mixed type. Shallow aquifer contains Mg–Na–HCO₃ and Ca–Mg–Na–HCO₃ facies and deep aquifer contains Ca–HCO₃ and Mg–HCO₃ Na–Cl facies [12].

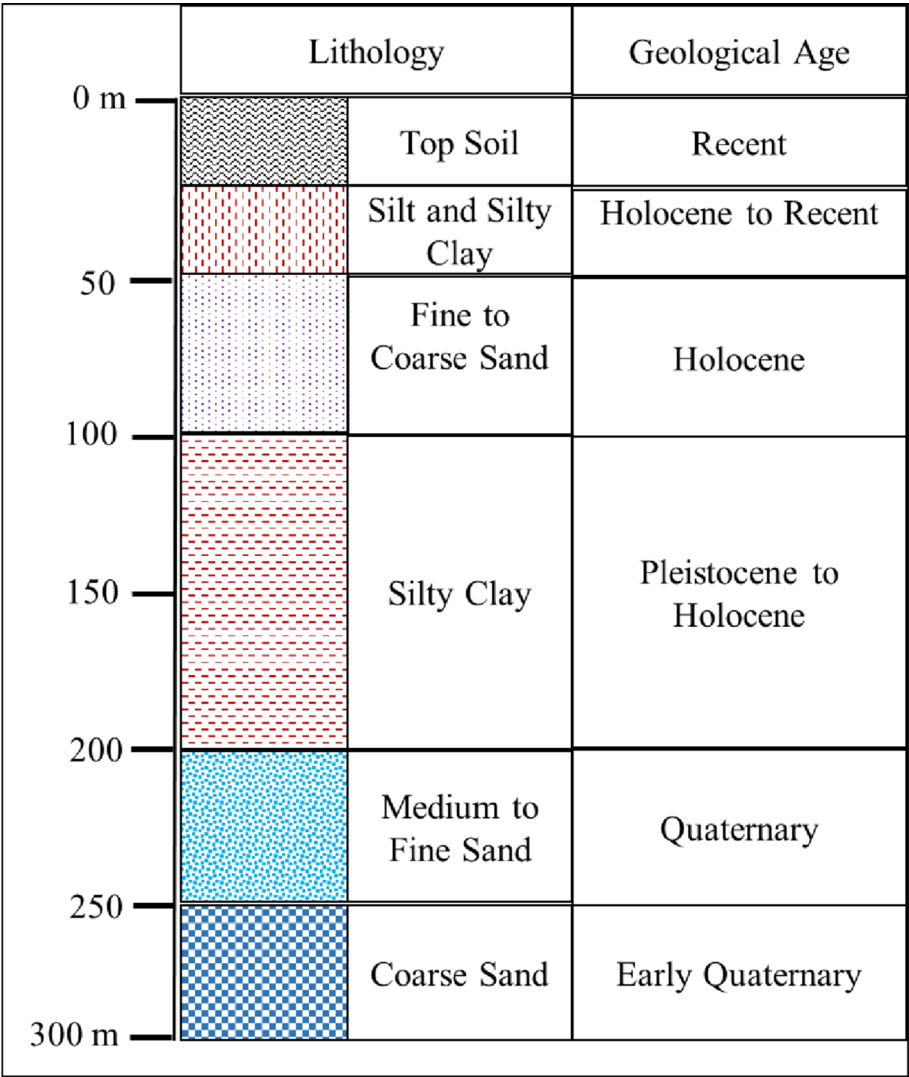


Fig. 2 Stratigraphy of the nearby Khulna city corporation (modified after Roy et al. [23])

3 Materials and methods

In recent decades, geophysical techniques have been widely applied in hydrogeological investigations because they provide non-invasive and cost-effective reliable subsurface information. Among these methods, vertical electrical sounding (VES) is particularly effective for identifying hydrostratigraphic layers and mapping subsurface resistivity variations associated with lithological changes [24–29].

The objective of this study was to characterize lithological heterogeneity and infer hydro-stratigraphic conditions in the floodplain of Jashore Sadar using vertical electrical sounding (VES). A total of 18 VES points were conducted across the study area to ensure adequate spatial representation of subsurface variability. The number and distribution of VES locations were selected to capture major geomorphological units and lithological transitions typical of alluvial floodplain environments, while also considering site accessibility and land-use constraints. The spacing between adjacent VES points ranged from approximately 1 to 5 km, which is consistent with previous hydrogeophysical

investigations in similar sedimentary settings [29–33]. The geographic coordinates of all VES points were recorded using a handheld GPS (accuracy ± 3 m).

The resistivity survey was conducted using the Schlumberger electrode configuration in accordance with standard procedures outlined in ASTM D6431 [34]. In this configuration, the current electrodes (A and B) and the potential electrodes (M and N) were arranged symmetrically along a straight profile to ensure reliable measurement of subsurface resistivity variations (Fig. 3a). To achieve the overall objectives, every step was carried out through a systematic sequence of operations (Fig. 3b). Initially, the VES locations were selected and marked based on site conditions and survey design. Prior to measurement, electrode contact resistance was carefully assessed and optimized to ensure good electrical coupling with the ground. Electrical current was then injected into the subsurface through the outer electrodes (A–B), while the resulting potential difference was measured across the inner electrodes (M–N). To probe deeper subsurface layers, the current electrode spacing ($AB/2$) was progressively increased. When the measured potential difference became too small to record accurately, the spacing between the potential electrodes ($MN/2$) was increased accordingly to maintain measurement sensitivity. At each electrode configuration, measurements were repeated and stacked to improve the signal-to-noise ratio and enhance data reliability.

The current electrode half-spacing ($AB/2$) varied from approximately 2 to 250 m, depending on site constraints and available space for electrode deployment. The potential electrode spacing ($MN/2$) was initially kept small, typically within the range of 0.5 to 2 m, and was increased as necessary during the survey. Based on the maximum current electrode spacing, the depth of investigation is estimated to be approximately 90 to 150 m, assuming a penetration depth of one-third to one-fifth of the total current electrode separation. All field measurements were performed using a McOHM Model-2115 resistivity meter (OYO Corporation, Tokyo, Japan). The field campaign was executed over a 7-day period from December 12 to December 18, 2016, during the dry pre-monsoon season to minimize soil moisture fluctuations. Throughout the survey, careful attention was given to minimizing sources of noise, including poor electrode grounding (contact resistance $<2k\Omega$) and potential interference from nearby cultural features, to ensure the accuracy and consistency of the acquired data. An automatic stacking cycle of 3 to 4 repetitions per data point was enforced, successfully lowering the standard deviation of individual voltage readings below 0.5 mV.

Quantitative interpretation of the VES data was performed using IX1D software (Interpex Ltd.), which applies a 1D layered earth inversion algorithm (Fig. 4). The initial models derived from curve matching were used as input for inversion. The IX1D inversion was configured with a least-squares smoothness-constrained algorithm using an

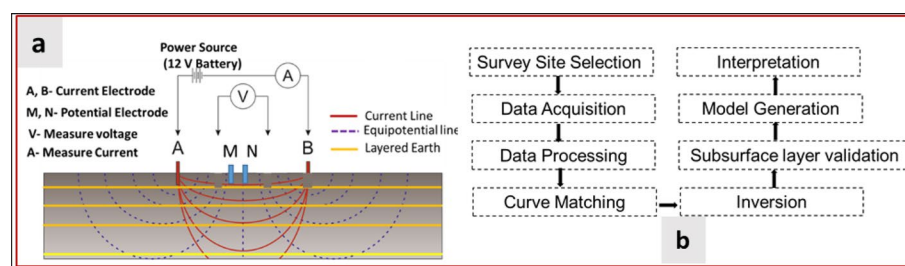


Fig. 3 (a) VES layout and (b) methodology

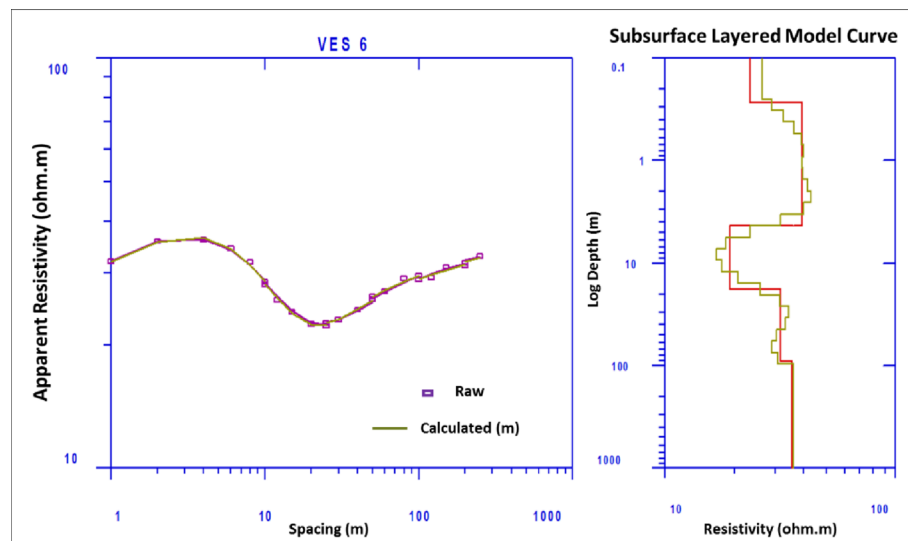


Fig. 4 A representative apparent resistivity well-fitted curve and inverse smooth and subsurface layered electric model of VES-6

initial damping factor of 0.5 and a layer thickness increment factor of 1.1. The Occam's inversion method was utilized with a uniform regularization parameter (λ) starting at 10 and a reduction factor of 2 per iteration. The number of layers in the model ranged from 4 to 5, depending on the complexity of the field curve. Convergence was defined as an RMS change $< 0.1\%$ between iterations, with a maximum of 10 iterations allowed. The quality of the model fit was evaluated using the root mean square (RMS) error between observed and calculated apparent resistivity values. The exact point-specific RMS error values, spatial coordinates, and individual layer counts for all 18 VES soundings are cataloged in Table 2. The errors ranged from a minimum of 1.31% (VES-10) to a maximum of 5.02% (VES-09), confirming a highly reliable fit across the entire study area.

Lithological interpretation of subsurface layers was performed by integrating resistivity data with borehole lithological information and established resistivity characteristics of unconsolidated alluvial sediments [35] (Fig. 5). A site-specific resistivity-based classification scheme was developed, in which low resistivity values are interpreted as clay-rich or saturated fine-grained materials, moderate values correspond to silty clay and fine sand, and high resistivity values represent medium to coarse sand and gravel units that are typically associated with aquifer zones.

As shown on Fig. 5, the known overlap of resistivity ranges due to variations in moisture content, pore water conductivity, and clay fraction, the interpretation was constrained using borehole lithology to reduce ambiguity arising from equivalence and suppression effects. To evaluate the reliability of the VES-derived models, three borehole logs were used for validation. The boreholes were located within 500–1000 m of selected VES points, enabling comparison of layer boundary depths, thicknesses, and lithological classifications. Using Google Colab python environment, these parameters were used to measure correlation coefficient between VES-derived and borehole-derived depths, thickness and accuracy of lithological classifications using confusion matrix. Finally, to observe the subsurface lithological heterogeneity, four two-dimensional geoelectric cross-sections oriented along north–south, east–west, northwest–southeast, and

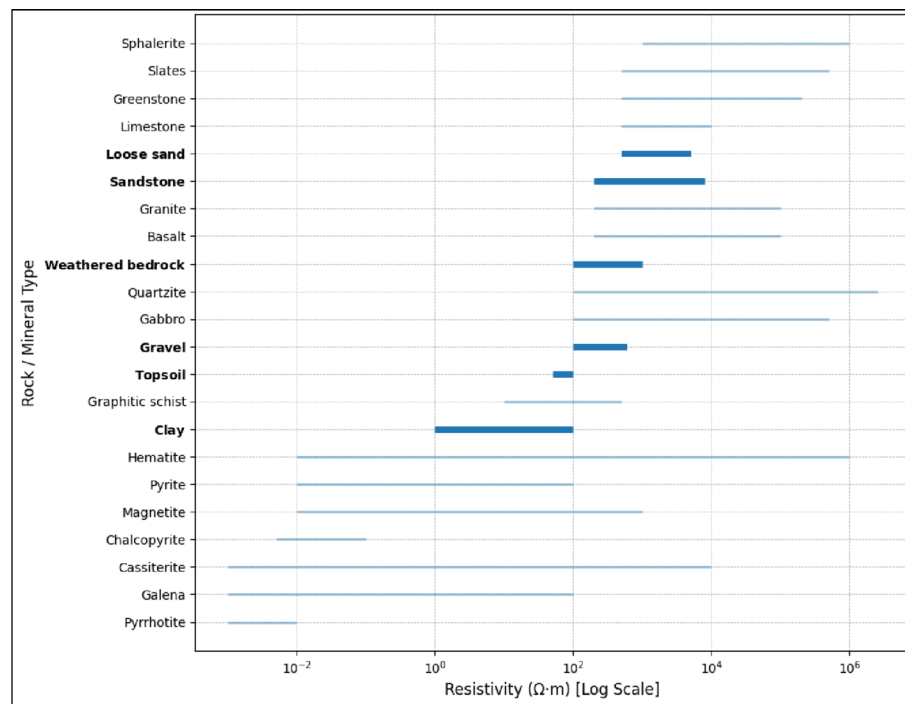


Fig. 5 The resistivity-lithology relationship of typical rocks and minerals while the resistivity of different alluvial sedimentary rocks is highlighted in bold in color in the image (modified after Octova and Yulhendra [35])

northeast–southwest directions were constructed and visualized using RockWorks 2015, while spatial mapping was performed in ArcGIS 10.5.

Nevertheless, it is important to note that the interpretation of VES data remains subject to inherent limitations. The non-uniqueness of geophysical inversion implies that multiple subsurface models can produce similar apparent resistivity responses, while equivalence and suppression effects may obscure thin or low-contrast layers. Furthermore, lateral heterogeneity in alluvial environments can violate the assumption of horizontal layering inherent in 1D inversion approaches. Additional uncertainties may arise from factors such as electrode contact resistance, environmental noise, and instrumental constraints, although these were minimized through careful field procedures and signal stacking. While integration with borehole data improves model reliability, some discrepancies are unavoidable due to spatial separation between VES points and borehole locations.

4 Results and discussion

All 18 soundings achieved inversion convergence with RMS errors ranging from 1.31% (VES-10) to 5.02% (VES-9) (Table 1). These RMS values are well within the acceptable threshold (< 10%) for 1D VES interpretation in alluvial environments [33], confirming that the 1D layered models provide reliable representations of the local subsurface resistivity structure.

Further, to evaluate the reliability of the VES-derived interpretations, the geoelectric models were compared layer-by-layer with three nearby borehole logs (Table 1). Quantitative validation using statistical measures shows strong correspondence between VES-derived and borehole-derived layer depths and thicknesses, with coefficients of determination (R^2) exceeding 0.95 and mean absolute error generally below 2.5 m

Table 1 Layer by layer comparison between VES and borehole lithology

VES ID	*BH ID	Layer No.	Depth VES (m)	Depth BH (m)	Thickness VES (m)	Thickness BH (m)	Lithology BH	Interpreted Lithology VES
VES-6	BH-1	1	0.7	2	0.7	2	Top Soil	Top Soil
VES-6	BH-1	2	4.7	6	4	4	Silty Clay	Silty Clay
VES-6	BH-1	3	23.7	20	16	14	Medium Sand	Fine sand
VES-6	BH-1	4	108.7	103.7	85	83.7	Silty Clay	Silty Clay
VES-9	BH-2	1	0.4	1	0.4	1	Top Soil	Top Soil
VES-9	BH-2	2	2.4	2	2	1	Silty Clay	Silty Clay
VES-9	BH-2	3	49.4	50	47	48	Fine Sand	Clay
VES-9	BH-2	4	149.4	150	100	100	Fine Sand	Fine Sand
VES-18	BH-3	1	0.5	2	0.5	2	Top Soil	Top Soil
VES-18	BH-3	2	3	6	2.5	4	Silty Clay	Fine Sand
VES-18	BH-3	3	38	40	35	34	Clay	Clay
VES-18	BH-3	4	108	100	70	60	Medium Sand	Medium Sand

*BH=Borehole

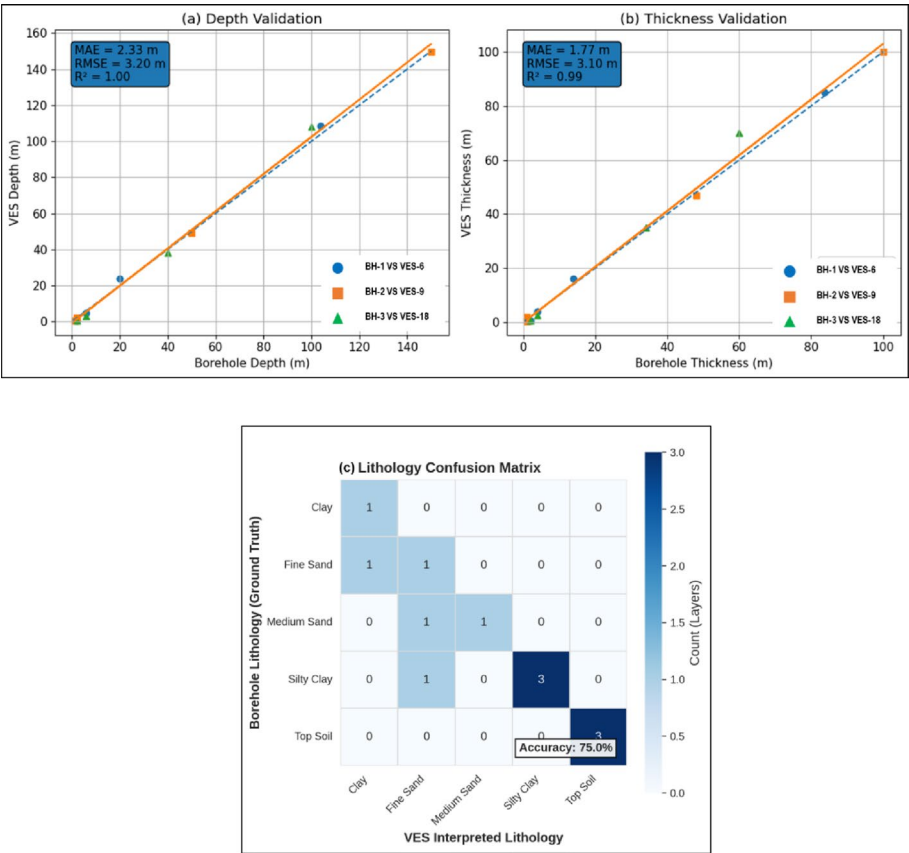


Fig. 6 Validation of lithological (a) depths, (b) thickness and (c) quantitative assessment of the accuracy of lithological classifications between VES and borehole

(Fig. 6a, b). In addition, comparison between borehole lithology and VES-derived lithological classification yields an overall agreement of approximately 75% (Fig. 6c). These results indicate a generally satisfactory consistency between interpreted resistivity layers and observed subsurface lithology, despite some local discrepancies. Such discrepancies are expected because resistivity inversion is inherently non-unique and because the borehole and VES locations are not spatially coincident. Nevertheless, while very few

studies use VES to interpret subsurface lithological heterogeneity regarding geological condition of an area due to ambiguity of overlapping resistivity, our study evaluate the comparative quantitative assessment of uncertainty with the bore logs using statistical measures to reduce the uncertainty and increase the reliability of the VES-derived models to enhance the understanding of subsurface lithological heterogeneity of quaternary alluvial floodplain environment.

Based on the validated VES interpretations, five distinct geoelectric layers were identified within the investigated depth range of approximately 90–150 m. These layers were broadly categorized as top soil, clay, silty clay, fine sand, and fine to medium grained sand units based on their resistivity characteristics, thickness variation, and correlation with available borehole information (Tables 2 and 3). The identified lithological succession is generally consistent with previous geological and hydrogeological studies conducted in southwestern Bangladesh, which describe heterogeneous Quaternary floodplain deposits composed predominantly of alternating sand, silt, and clay layers [17, 36].

To examine the spatial distribution of these units, four two-dimensional geoelectric cross-sections oriented along north–south, east–west, northwest–southeast, and north–east–southwest directions were constructed (Fig. 7a and b). The cross-sectional profiles reveal substantial variability in resistivity, thickness, and depth of subsurface layers across the study area, reflecting the frequent facies changes and heterogeneous nature of Quaternary floodplain deposits. For example, lateral continuity of sand up to 110 m depth is often truncated by clay or silty clay unit due to the floodplain fluvial deposits. These findings are corroborated by different researchers around the world who found similar pattern of lithological heterogeneity in alluvial sedimentary flood plain environment studying potential groundwater development [20, 24, 28, 37, 38].

The first geoelectric unit, interpreted as top soil, exhibits resistivity values ranging from 6 to 30.5 Ωm and thickness varying between 0.25 m and 2 m (Table 2). Cross-sectional profiles suggest that this unit is comparatively thinner in the north-central part of the study area (VES-18, VES-17, VES-11, VES-4, VES-7, VES-10, and VES-14). The observed variation in resistivity likely reflects differences in moisture content, grain-size composition, and degree of sediment compaction [21]. Although relatively thin, this unit may influence localized recharge and near-surface infiltration processes. However, due to its limited thickness and predominantly fine-grained composition, it is unlikely to contribute substantially to regional groundwater storage.

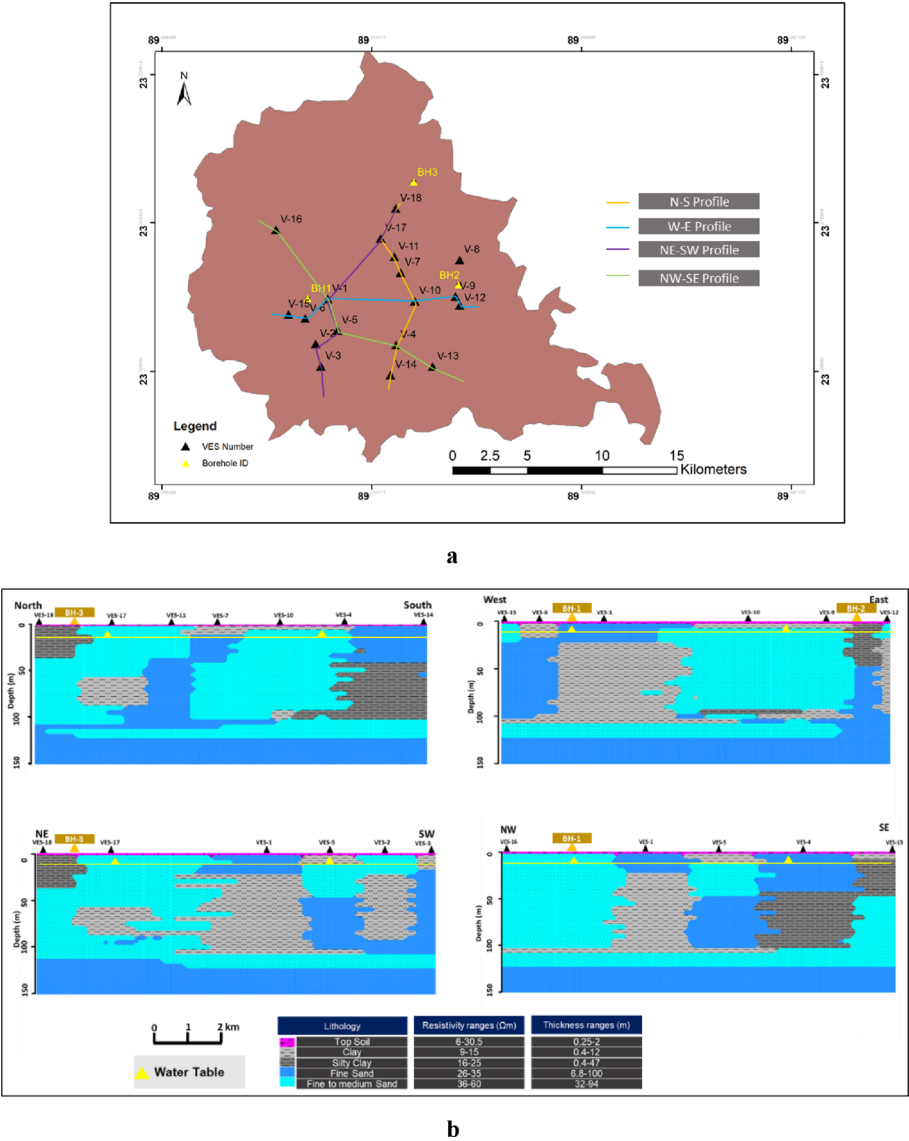
The second geoelectric unit, interpreted as clay, shows relatively low resistivity values ranging from 9 to 15 Ωm and thickness varying between 0.4 m and 12 m. Spatially, this unit appears discontinuous or locally absent in several locations, including VES-4, VES-14, and VES-18. The third geoelectric unit, interpreted as silty clay, demonstrates resistivity values ranging from 16 to 25 Ωm and thickness varying between 0.4 m and 47 m. Similar to the clay layer, the unit displays marked lateral variability and occurred either beneath the clay layer or directly below the top soil in some areas, indicating localized variation in depositional conditions. The relatively low to moderate resistivity values observed in these units are generally consistent with fine-grained sediments having comparatively lower permeability and higher moisture retention [38]. Consequently, these units may function hydrostratigraphically as confining or semi-confining layers (aquitards). However, our interpretation is based primarily on lithological inference from resistivity data and is not supported by direct hydraulic measurements such as pumping tests

Table 2 Characteristics of subsurface geo-electric layers obtained from 18 VES at Jashore Sadar Upazila

VES No.	Long	Lat	Layers resistivity (ohm-m)						Layers thickness (m)					Total thickness	RMS error (%)	Res layers no.
			ρ1	ρ2	ρ3	ρ4	ρ5	ρ6	h1	h2	h3	h4	h5			
VES-01	89.188306	23.169528	28	22	32	22	38	-	0.7	4	16	85	-	105.7	4.77	5
VES-02	89.180917	23.142472	16.9	12	40	24	36	-	0.6	3.9	16.1	74	-	94.6	3.65	5
VES-03	89.184139	23.128972	13.3	10	16	29	26	-	0.4	0.9	18	86	-	105.3	3.27	5
VES-04	89.229306	23.141833	22	26	20	25.7	15	37	0.25	0.57	4.01	40	56	100.83	2.16	6
VES-05	89.194139	23.150528	10	9	21	39	28	33	0.8	1.8	8.93	34	55	100.53	1.73	6
VES-06	89.174722	23.157778	23.5	39.5	19	32	36	-	0.25	4.05	14	78	-	96.3	2.36	5
VES-07	89.232056	23.185333	30.5	27.5	22	32	37	35	0.35	2.55	8	34	56	100.9	2.03	6
VES-08	89.267667	23.193056	29	10	32	23	40	-	2	12	24	64	-	102	2.88	5
VES-09	89.265	23.171	18	21	14	30	35	-	0.4	2	47	100	-	149.4	5.02	5
VES-10	89.240583	23.16825	30	40	25	43	34.2	-	0.3	1.4	3.5	93	-	110	1.31	5
VES-11	89.228278	23.194944	15	22	48	28	39	-	0.5	1.3	34.2	73	-	109	1.60	5
VES-12	89.267667	23.16575	25	17	40	22	28	-	0.25	0.6	17.15	77	-	95	2.86	5
VES-13	89.251389	23.128861	22	32	18	9	36	24	0.25	1	5	40	74	120.25	2.74	6
VES-14	89.226061	23.123669	6	25	9	34	25	-	0.9	5.7	21	77	-	103.6	2.99	5
VES-15	89.164611	23.160194	19	22	42.5	33	31	-	0.4	0.4	19.2	82	-	102	1.33	5
VES-16	89.157083	23.211222	21	22	59	52	45	42	0.4	0.4	3.2	6.8	94	104.8	2.34	6
VES-17	89.220083	23.206083	18.4	31.25	18.9	41.3	21.7	40	0.67	1.4	0.4	54.4	32.35	93.5	1.92	6
VES-18	89.229139	23.224083	19	28	11	36	27	-	0.5	2.5	35	70	-	108	2.0	5
Min.			6	9	9	9	15	24	0.25	0.4	0.4	6.8	32.35			
Max.			30.5	40	59	52	45	42	2	12	47	100	94			
Mean			20.37	23.13	27.08	30.83	32.11	35.17	0.55	2.58	16.37	64.90	61.23			
Std.Dev.			6.60	9.03	13.51	9.42	7.24	5.81	0.40	2.71	12.25	24.08	18.99			

Table 3 Thickness range and corresponding resistivity values of the geo-electric layers

Geo-electric layers	Thickness ranges (m)	Range of corresponding resistivity (Ω.m)
Top soil	0.25–2	6–30.5
Clay	0.4–12	9–15
Silty clay	0.4–47	16–25
Fine sand	6.8–100	26–35
Fine to medium sand	32.35–94	36–60



or permeability analyses. Furthermore, because the lateral continuity of these units was interpreted from stitched one-dimensional VES models, their regional continuity should be regarded as indicative rather than definitive.

The fourth geoelectric unit, fine sand, is characterized by resistivity values ranging from 26 to 35 Ωm and thickness varying between 6.8 m and 100 m. This unit appears relatively widespread throughout the study area. In some locations (e.g., VES-1 and VES-14), it directly underlies the top soil, whereas in other areas it occurs beneath intervening clay and silty clay deposits. The fifth geoelectric unit, fine to medium grained sand, indicate comparatively higher resistivity values ranging from 36 to 60 Ωm and thickness between 32.35 m and 94 m. This unit also appears laterally extensive and, in several locations (e.g., VES-2, VES-16, and VES-17), occurs immediately beneath the top soil, indicating notable spatial variation in stratigraphic succession.

The comparatively higher resistivity values and substantial thicknesses of the fourth and fifth geoelectric units suggest relatively more permeable subsurface formations, and indicates potential aquifer zones. Nevertheless, it is important to emphasize that VES data alone cannot directly determine aquifer productivity, transmissivity, groundwater yield, or groundwater quality. The interpretation of these units as aquifers is therefore based on inferred lithological characteristics rather than direct hydrogeological testing. Confirmation of aquifer potential would require complementary investigations including pumping tests, hydraulic conductivity measurements, and hydrochemical analyses.

For hydrostratigraphic interpretation, the five identified geoelectric layers may be grouped into three broader hydrostratigraphic units according to their inferred hydrogeological behavior. The first hydrostratigraphic unit corresponds to the near-surface layer composed predominantly of silty sediments with minor clay and sand fractions identified from borehole logs. Although relatively thin, this unit appears laterally continuous across much of the study area. The second hydrostratigraphic unit comprises the clay and silty clay geoelectric layers, which together may function as a confining or semi-confining unit due to their inferred low permeability. This unit appears to become comparatively thicker toward the southeastern part of the study area, suggesting spatial variability in depositional processes and sediment accumulation. The third hydrostratigraphic unit consists of the fine sand and fine to medium grained sand layers, which together represent the principal potential aquifer zones identified in the study area.

Compared with previous investigations in southwestern Bangladesh, which relied mainly on sparse borehole information or hydrochemical facies analysis [14, 16, 17], the present study provides a more spatially continuous interpretation of subsurface heterogeneity using VES. Earlier studies documented the general occurrence of sandy aquifers interbedded with clay-rich deposits but provided limited information regarding lateral continuity, thickness variation, and subsurface geometry. By integrating VES interpretation with available borehole data and conducting quantitative validation, the present study improves understanding of the vertical and lateral heterogeneity of Quaternary floodplain deposits in Jashore Sadar.

From groundwater management perspectives, the observed lithological heterogeneity has important implications for aquifer targeting and well placement. Variations in the depth, thickness, and continuity of sandy units suggest that groundwater development strategies should be site-specific rather than generalized across the entire area. For example, locations where sandy units occur at relatively shallow depths (VES-4, 9, 10,

11, 14, 15, 16, 17) or exhibit greater thickness (VES-3, 5, 6, 7, 10) may represent comparatively favorable zones for groundwater abstraction whereas areas dominated by thick clay-rich deposits ((VES-1, 12, 13, 18) may exhibit lower recharge potential and reduced well productivity. Therefore, the spatial patterns identified from the VES cross-sections may provide a preliminary framework for groundwater exploration and abstraction planning. However, all proposed development sites should be validated using borehole investigations and hydraulic testing prior to large-scale groundwater abstraction.

Interestingly, an additional observation from our study is the absence of extremely low resistivity signatures typically associated with saline water intrusion. This contrasts with neighboring coastal and southwestern districts such as Khulna, Satkhira, Barguna and Bagerhat, where saline intrusion has become a major hydrogeological concern [39–42]. Although resistivity data alone cannot conclusively confirm groundwater freshness, the observed resistivity distribution suggests that significant saline intrusion is unlikely within the investigated depth range of the study area. However, hydrochemical validation should be necessary to confirm groundwater quality conditions.

Despite its contributions, the present study has several limitations. However, the limitations can open up new windows for further research. For example, the interpretation of subsurface lithology is based primarily on resistivity data, which is inherently non-unique and may correspond to multiple lithological scenarios. Second, the absence of pumping-test data limits the ability to quantify hydraulic properties such as transmissivity and hydraulic conductivity. Third, hydrochemical analyses were not incorporated; therefore, no direct conclusions can be drawn regarding groundwater quality or suitability for drinking purposes. Additionally, the use of one-dimensional VES data to construct lateral interpretations introduces uncertainty in representing complex three-dimensional floodplain heterogeneity. Seasonal variations in groundwater saturation conditions were also not explicitly considered, although such variations may influence resistivity responses. Finally, the use of only 18 VES points limits the spatial resolution of the interpreted cross-sections and may not fully represent heterogeneity across the entire Jashore Sadar area.

Future studies should therefore incorporate denser geophysical coverage, pumping tests, borehole verification, and hydrochemical analyses to improve characterization of aquifer geometry, hydraulic properties, and groundwater quality. Integration of geophysical, hydrogeological, and hydrochemical datasets would provide a more comprehensive framework for potential groundwater resource assessment and management in complex floodplain environments. Nonetheless, our study provides an important conceptual framework of lithological heterogeneity using VES which has substantial implications on potential groundwater management in the floodplain of Jashore Sadar and similar floodplain environment around the world.

5 Conclusions

The research highlights the capability of vertical electrical sounding (VES) to characterize the subsurface lithological heterogeneity and its implication on potential groundwater development in the active floodplain of the Quaternary deposits of the Jashore area. The analysis after validation with nearby borelogs identified five geoelectric layers based on variations in resistivity (6–60 Ωm) and thickness. The results reveal significant lateral and vertical heterogeneity in subsurface layering, particularly in the distribution and

thickness of clay-rich and sandy units, indicating pronounced facies variations across the study area. Geo-electric cross-sections along four profiles (N–S, E–W, NW–SE, and NE–SW) further highlight these heterogeneities, with shallow or thick sandy units occurring at VES-3–7, 9–11, and 14–17, whereas clay and silty clay dominated sequences are prominent at VES-1, 12, 13, and 18. The deeper sandy layers, characterized by relatively high resistivity (36–60 Ω m) and substantial thickness (32.35–94 m), are interpreted as potential aquifer zones, while the overlying clay and silty clay layers likely act as confining to semi-confining units. In fact, the spatial variability in thickness and depth of these units suggests that groundwater occurrence and accessibility are highly site-dependent across the study area. Interestingly, the absence of very low resistivity signatures indicates limited saline water intrusion within the investigated depth range compared to nearby areas. However, these interpretations are based on geophysical responses and lithological inference, and therefore should be considered indicative rather than definitive. Importantly, VES data alone do not provide direct information on groundwater quality, transmissivity, or well yield. As such, the identification of aquifer potential in this study reflects relative suitability based on subsurface characteristics, rather than confirmed groundwater productivity or water quality conditions. Despite these limitations, the study provides a spatially continuous framework of subsurface variability, which can support potential groundwater management and guide future groundwater exploration efforts. The findings are particularly useful for identifying zones where further investigation—such as borehole logging, pumping tests, and hydrochemical analysis—should be prioritized. Overall, this study contributes to improving the understanding of subsurface heterogeneity in floodplain environments and highlights the value of integrating geophysical methods with hydrogeological data for more reliable groundwater assessment and management.

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Author contributions

M.I.S.H.: Conceptualization, Field investigation, Formal analysis, Original manuscript writing, Project administration, Methodology, Data curation, Visualization, Validation, Software; M.M.R.: Formal analysis, Original manuscript writing, Methodology, Validation; M.G.M.: Reviewing and editing. M.S.R.: Manuscript editing, Map preparation. A.S.M.W.: Conceptualization, Field investigation, Formal analysis, Project administration, Methodology, Visualization, Validation.

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Data availability

The datasets used and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

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Competing interests

The authors declare no competing interests.

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References

1. Loaiciga HA, Doh R. Groundwater for people and the environment: a globally threatened resource. 2023. <https://doi.org/10.1111/gwat.13376>.
2. Jasechko S, et al. Rapid groundwater decline and some cases of recovery in aquifers globally. *Nature*. 2024;625(7996):715–21. <https://doi.org/10.1038/s41586-023-06879-8>.
3. Khan MYA, et al. Influence of natural and anthropogenic factors on the hydrogeology and hydrogeochemistry of Wadi Itwad Aquifer, Saudi Arabia: Assessment using multivariate statistics and PMWIN simulation. *Ecol Indic*. 2023;151:110287.
4. Khalifa M, et al. Evaluation of groundwater resources in Wadi Qena, Egypt: a geophysical and hydrogeochemical perspective. *Sci Rep*. 2025;15(1):43834–. <https://doi.org/10.1038/s41598-025-29240-7>. Dec. 2025.
5. ElKashouty M, Khan MYA, Alharbi K, Pande CB, Subyani AM, Tian F. Hydrogeology and hydrogeochemistry of saline groundwater seepage zones in Wadi Bani Malik basin, Jeddah, Saudi Arabia: Impacts on soil and water resources. *Water*. 2023;15(19):3464.
6. Shamsudduha M, Joseph G, Haque SS, Khan MR, Zahid A, Ahmed KMU. Multi-Hazard Groundwater Risks to the Drinking Water Supply in Bangladesh: Challenges to Achieving the Sustainable Development Goals, *Multi-Hazard Groundw. Risks to Drink. Water Supply Bangladesh Challenges to Achiev. Sustain. Dev. Goals*. no June. 2019. <https://doi.org/10.1596/1813-9450-8922>.
7. Shamsudduha M, Joseph G, Haque SS, Khan MR, Zahid A, Ahmed KMU. Multi-hazard groundwater risks to water supply from shallow depths: challenges to achieving the sustainable development goals in Bangladesh. *Expo. Heal*. 2020;12(4):657–670. <https://doi.org/10.1007/s12403-019-00325-9>.
8. Islam MS, Mostafa MG. Groundwater status and challenges in Bangladesh. 2021; 79–146. https://doi.org/10.1007/978-3-030-73245-5_4.
9. Murshed SB, Kaluarachchi JJ. Scarcity of fresh water resources in the Ganges Delta of Bangladesh. *Water Secur*. 2018;4–5. <https://doi.org/10.1016/j.wasec.2018.11.002>.
10. Shahid S, Hazarika MK. Groundwater drought in the northwestern districts of Bangladesh. *Water Resour Manag*. 2010;24(10):1989–2006. <https://doi.org/10.1007/s11269-009-9534-y>.
11. Mojid MA, Parvez MF, Mainuddin M, Hodgson G. Water table trend-a sustainability status of groundwater development in North-West Bangladesh. *Water (Switzerland)*. 2019;11(6). <https://doi.org/10.3390/w11061182>.
12. Dola SS, Bahsar K, Islam M, Sarker MMR. Hydrogeological Classification and the Correlation of Groundwater Chemistry with Basin Flow in the South-Western Part of Bangladesh. *J Bangladesh Acad Sci*. 2018;42(1):41–54. <https://doi.org/10.3329/jbas.v42i1.37831>.
13. Ahmed MT, et al. Evaluation of hydrochemical properties and groundwater suitability for irrigation uses in southwestern zones of Jashore, Bangladesh. *Groundw Sustain Dev*. 2020;11:100441. <https://doi.org/10.1016/j.gsd.2020.100441>.
14. Chakraborty TK, et al. Human health risk and hydro-geochemical appraisal of groundwater in the southwest part of Bangladesh using GIS, water quality indices, and multivariate statistical approaches. *Toxin Rev*. 2023;42(1):285–99.
15. Chakraborty TK, et al. Arsenic, iron, and manganese in groundwater and its associated human health risk assessment in the rural area of Jashore, Bangladesh. *J Water Health*. 2022;20(6):888–902.
16. Rahman MM, et al. Hydrogeochemical facies analysis of groundwater at parts of south-western Bangladesh: a case study. *Arab J Geosci*. 2023;16(12):644.
17. Zahid A, et al. The aquifer mapping and nomenclature guidelines for the Bengal Basin, Bangladesh. *Environ Earth Sci*. 2026;85(5):119.
18. Ahmed M, Woobaidullah A, Quamruzzaman C, Hossain R, Islam MS, Mehnaz N. The evaluation of hydrogeological condition: a case study on Jessore Sadar. *Bangladesh Int J Emerg Technol Adv Eng*. 2019;9(6):17–26. https://www.researchgate.net/publication/334094605_The_Evaluation_of_Hydrogeological_Condition_A_Case_Study_on_Jessore_Sadar_Bangladesh.
19. Kamaraj P, Jothamani M, Panda B, Sabarathinam C. Mapping of groundwater potential zones by integrating remote sensing, geophysics, GIS, and AHP in a hard rock terrain. *Urban Clim*. 2023;51:101610.
20. Basharat U, Zhang W, Abbasi A, Mahroof S, Niaz A, Khan SH. Geoinformatics and hydrogeophysical-based delineation of groundwater potential zone through surface and subsurface indicators. *Appl Water Sci*. 2025;15(8):222.
21. Hossain MIS, et al. Integrated satellite imagery and electrical resistivity analysis of underground mine-induced subsidence and associated risk assessment of Barapukuria coal mine, Bangladesh. *Environ earth Sci*. 2023;82(22):537.
22. Alam M, Alam MM, Curry JR, Chowdhury MLR, Gani MR. An overview of the sedimentary geology of the Bengal Basin in relation to the regional tectonic framework and basin-fill history. *Sediment Geol*. 2003;155:3–4. [https://doi.org/10.1016/S0037-0738\(02\)00180-X](https://doi.org/10.1016/S0037-0738(02)00180-X).
23. Roy MK, Datta DK, Adhikari DK, Chowdhury BK, Roy PJ. Geology of the Khulna City Corporation. *J Life Earth Sci*. 2005;1(July):57–63.
24. Obasi P, Ekinya A, Akpa C, Edene E. Characterization of subsurface lithology and aquifer parameters using vertical electrical sounding (VES) for groundwater development in Igbo-Imabana, Southern Nigeria. *Adv Geol Geotech Eng Res*. 2022;4(3):12–31.
25. Nyaberi DM. Application of vertical electrical sounding in mapping lateral and vertical changes in the subsurface lithologies: a case study of Olbanita, Menengai Area, Nakuru, Kenya. *Open J Geol*. 2023;13(1):23–50.
26. Emberga TT, Opara A, Osutor A, Ogah F, Epuehie E. An Assessment of Geoelectric Layers for Groundwater Availability in Makurdi Local Government Area, Benue State, Nigeria. *Niger J Phys*. 2026;35(2):136–49.
27. Abishek SR, et al. Integrated assessment of soil and rock penetration across the crystalline basement of the lower Thamirabarani river basin, southern India. *Discov Civ Eng*. 2026;3(1):68.
28. Nyaberi DM. Use of Vertical Electrical Sounding in Mapping Lateral and Vertical Changes in Subsurface Lithologies: A Case Study of Olbanita, Menengai Area, Nakuru, Kenya. *Res Adv Environ Geogr Earth Sci Vol*. 2024;6:1–34.
29. Gaikwad S, Pawar NJ, Bedse P, Wagh V, Kadam A. Delineation of groundwater potential zones using vertical electrical sounding (VES) in a complex bedrock geological setting of the West Coast of India. *Model Earth Syst Environ*. 2022;8(2):2233–47.
30. Ashraf A, Ali M. Prospecting groundwater potential using vertical electrical sounding technique for sustainable agriculture in the Upper Indus Basin, Pakistan. *J Mt Sci*. 2025;22(11):4012–23.

31. Vijayaprabhu S, Aravindan S, Kalaivanan K, Venkatesan S, Ravi R. Groundwater investigation through vertical electrical sounding: a case study from southwest Neyveli Basin, Tamil Nadu. *Int J Energy Water Resour.* 2024;8(1):17–34.
32. Chibuike A, Chukwu AC, Kelechi OK. Efficiency and limitation of vertical electrical sounding in evaluation of groundwater potential in fractured shale terrain: a case study of Abakaliki Area Lower Benue Trough Nigeria. *Environ Monit Assess.* 2023;195(1):158.
33. Mele M, Bersezio R, Giudici M. Hydrogeophysical imaging of alluvial aquifers: electrostratigraphic units in the quaternary Po alluvial plain (Italy). *Int J Earth Sci.* 2012;101(7):2005–25. <https://doi.org/10.1007/s00531-012-0754-7>.
34. Guide for Using the Direct Current Resistivity Method for Subsurface Investigation,. May. 2010. <https://doi.org/10.1520/D6431-99R10>.
35. Octova A, Yulhendra D. Iron ore deposits model using geoelectrical resistivity method with dipole-dipole array. *MATEC Web Conf.* 2017;101:4017. <https://doi.org/10.1051/mateconf/201710104017>.
36. Islam MS, Amin MR, Haq FU. Geo-Electrical Resistivity Survey for Freshwater Aquifer Delineation: A Case Study in Coastal Areas of Barguna District, Bangladesh. *Water Conserv Sci Eng.* 2026;11(1):43.
37. Dars R, Qureshi AL, Jamali MA, Memon HAS, Kori SM, Oad S. Subsurface groundwater aquifer mapping and quality characterization in Matiari district, Sindh, Pakistan. *Environ Monit Assess.* 2023;195(1):22.
38. Lan X-X et al. Electrical resistivity tomography visualization of core-scale evolution to reveal the impact of lithologic heterogeneity on gas hydrate formation in marine sediments. *Pet Sci.* 2026. <https://doi.org/10.1016/j.petsci.2026.04.034>
39. Hossain I, Reza S, Shafiuzzaman SM, Sultan-UI-Islam M. Effect of seawater intrusion on water quality of coastal aquifer of Bagerhat district, Bangladesh. *Int J Energy Water Resour.* 2024;8(1):123–40.
40. Shaibur MR, Parvin S, Ahmmed I, Rahaman MH, Sarwar S. Climate change and salinity intrusion in the water sources of coastal Khulna district, Bangladesh. *Water-energy-nexus in the ecological transition: natural-based solutions, advanced technologies and best practices for environmental sustainability.* Springer; 2022. pp. 123–6. https://doi.org/10.1007/978-3-031-00808-5_29
41. Reza AHMS, Roy R, Barman T, Sarker PC, Khan AS. Evaluating groundwater vulnerability to seawater intrusion in the coastal region of Bangladesh. *Environ Nanotechnol Monit Manag.* Volume 25, 2026;101130. <https://doi.org/10.1016/j.enmm.2026.101130>
42. Akter T, Hoque MA-A, Mukul SA, Pradhan B. Coastal flood induced salinity intrusion risk assessment using a spatial multi-criteria approach in the South-Western Bangladesh. *Earth Syst Environ.* 2025;9(1):31–49.

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